IT5020
Social Network Analysis

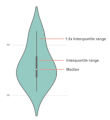
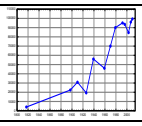
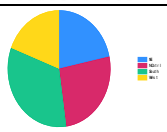
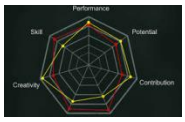
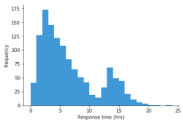
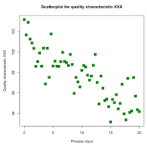
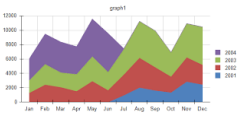
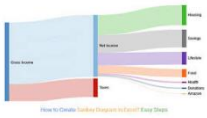
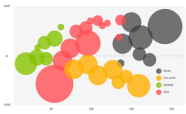
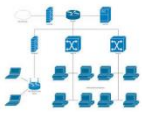
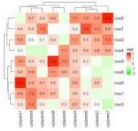
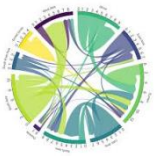
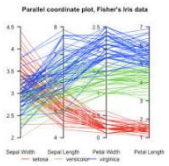
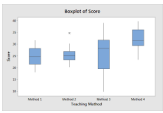
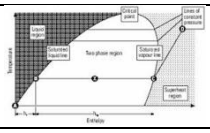
Assignment – II

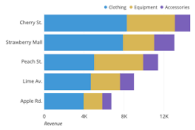
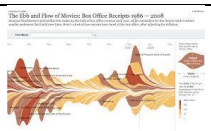
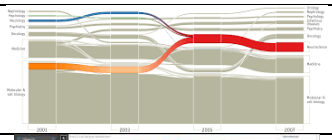
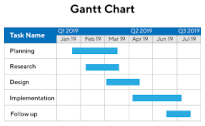
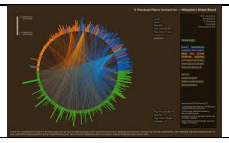
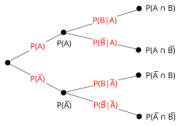
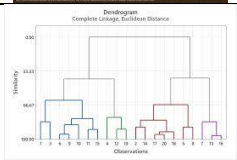
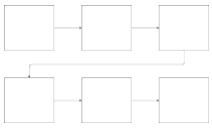
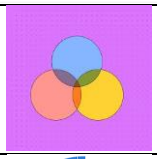
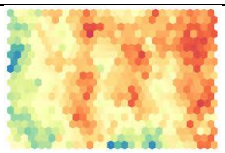
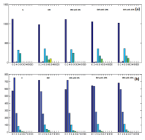
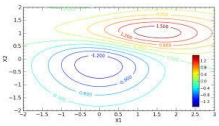
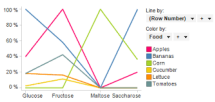
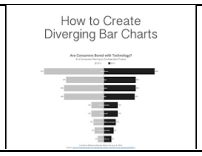
Name	Register No.	Contributions
Yuvaraj K	2022506079	Graph Visualisations, Ontology Development
Anishvaran G	2022506085	Community Detection and Search Algorithms
Krishnaa S	2022506104	Visualisation Representations, Graph Visualisations

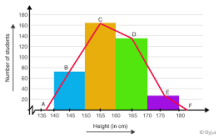
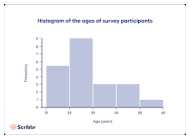
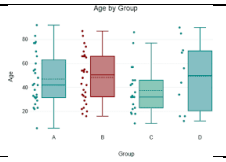
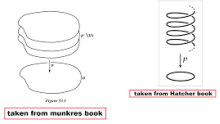
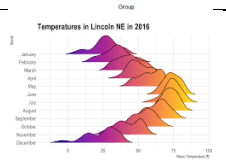
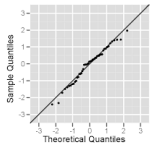
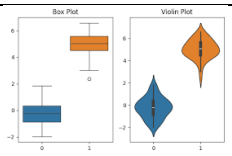
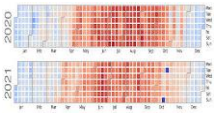
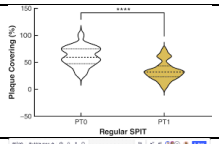
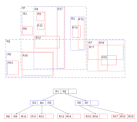
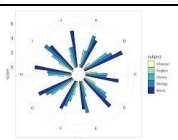
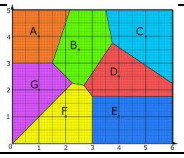
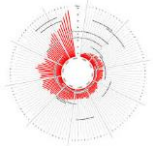
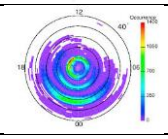
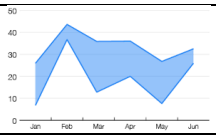
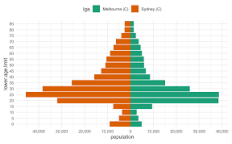
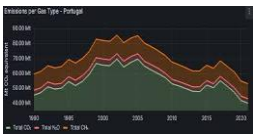
1: Community Detection and Search Algorithms

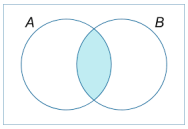
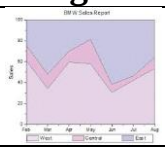
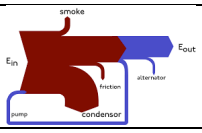
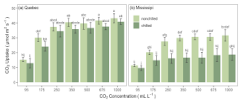
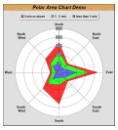
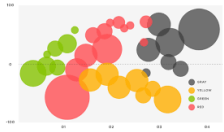
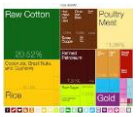
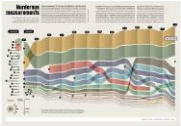
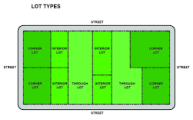
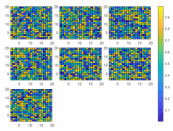
S. No.	Algorithm Name
1	Girvan–Newman Algorithm
2	Louvain Method
3	Label Propagation Algorithm (LPA)
4	Infomap
5	Walktrap
6	Fast Greedy Modularity Optimization
7	Edge Betweenness
8	Spectral Clustering
9	Hierarchical Clustering (Agglomerative)
10	K-Clique Communities
11	Clique Percolation Method (CPM)
12	OSLOM (Order Statistics Local Optimization Method)
13	CONGO (COmplex Network Graph Overlapping)
14	Leiden Algorithm
15	CESNA (Community detection via node attributes)
16	BigCLAM (Cluster Affiliation Model)
17	Modularity-Based Search
18	Random Walk Based Detection
19	SBM (Stochastic Block Model)
20	DeepWalk
21	Node2Vec
22	LINE (Large-scale Information Network Embedding)
23	Attractor Algorithm
24	Graph Autoencoder
25	Louvain-CPM Hybrid

2: List of Visualisations

S. No.	Visualization	Diagram	S. No.	Visualization	Diagram
1	Bar Chart		11	Violin Plot	
2	Line Chart		12	Donut Chart	
3	Pie Chart		13	Radar Chart	
4	Histogram		14	Polar Area chart	
5	Scatter Plot		15	Waterfall Chart	
6	Area Chart		16	Sankey Diagram	
7	Bubble Chart		17	Network Diagram	
8	Heatmap		18	Chord Diagram	
9	Treemap		19	Parallel Coordinates	
10	Box Plot		20	Steam Plot	

S. No.	Visualization	Diagram	S. No.	Visualization	Diagram
21	Stacked Bar Chart		31	Network Graph	
22	Streamgraph		32	Alluvial Diagram	
23	Gantt Chart		33	Word Cloud	
24	Flow Diagram		34	Map Chart	
25	Chord plot		35	Tree Diagram	
26	Dendrogram		36	Flow Map	
27	Venn Diagram		37	Hexagonal Heatmap	
28	Sunburst Chart		38	Histogram Heatmap	
29	Circular Packing		39	Contour Plot	
30	Parallel Coordinates		40	Diverging Bar Chart	

S. No.	Visualization	Diagram	S. No.	Visualization	Diagram
41	Frequency Polygon		51	Frequency Distribution	
42	Boxplot		52	Munkres Diagram	
43	Ridge Plot		53	Probability Plot	
44	Violin Box Plot		54	Time-Series Heatmap	
45	Split Violin Plot		55	Geographic Heatmap	
46	Bubble Map		56	R-Trees Chart	
47	Radial Bar Chart		57	Voronoi Diagram	
48	Circular Histogram		58	Bubble Matrix	
49	Polar Histogram		59	Area Range Chart	
50	Pyramid Plot		60	Time-Series Stacked Area	

S. No.	Visualization	Diagram	S. No.	Visualization	Diagram
61	Venn Diagram		71	Stacked Area Chart	
62	Sankey Diagram		72	Bar Plot	
63	Polar Area Chart		73	Bubble Chart	
64	TreeMap		74	Zoning	
65	Streamgraph		75	Line Lot	
66	Heat Map		76	Tile Plot	
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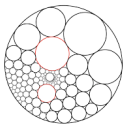
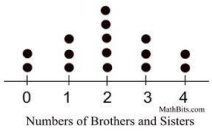
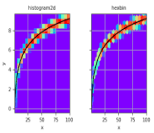
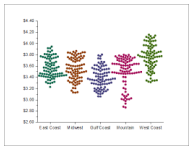
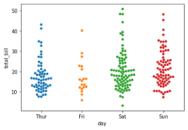
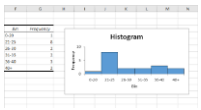
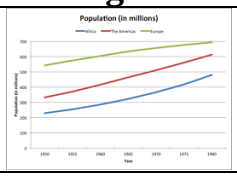
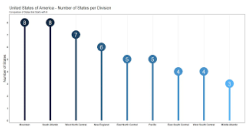
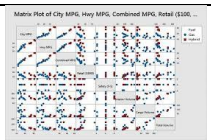
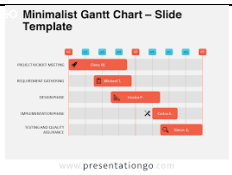
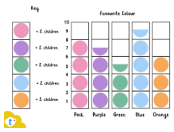
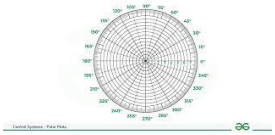
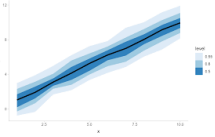
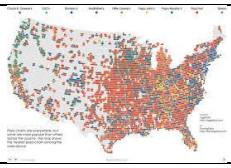
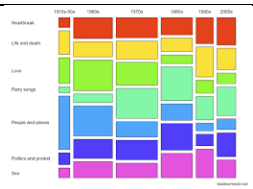
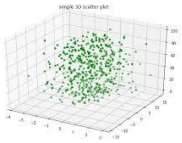
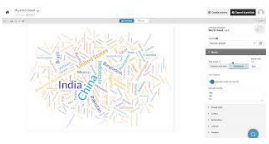
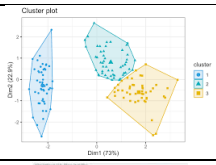
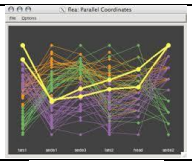
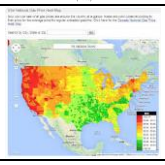
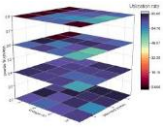
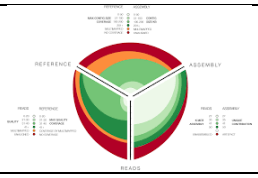
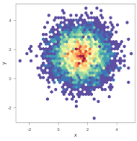
68	Circle Packing		78	Dot Plot	
69	Histogram2D		79	Bee Swarm Plot	
70	Swarm Plot		80	Histogram	

S. No.	Visualization	Diagram	S. No.	Visualization	Diagram
81	Multiple Line Plot		91	Network Graph	
82	Lollipop Chart		92	Matrix Plot	
83	Gantt Chart		93	Pictogram	
84	Polar Plot		94	Circular Barplot	
85	Ribbon Plot		95	Dot Density Plot	
86	Mosaic Diagram		96	3D Scatter Plot	
87	Wordcloud		97	K-Means Cluster Plot	
88	Parallel Coordinates		98	Geospatial Heatmap	
89	Spiral Plot		99	3D Heatmap	
90	Hive Plot		100	Hexbin Plot	

3: Graph Visualisations

High-Tech Managers

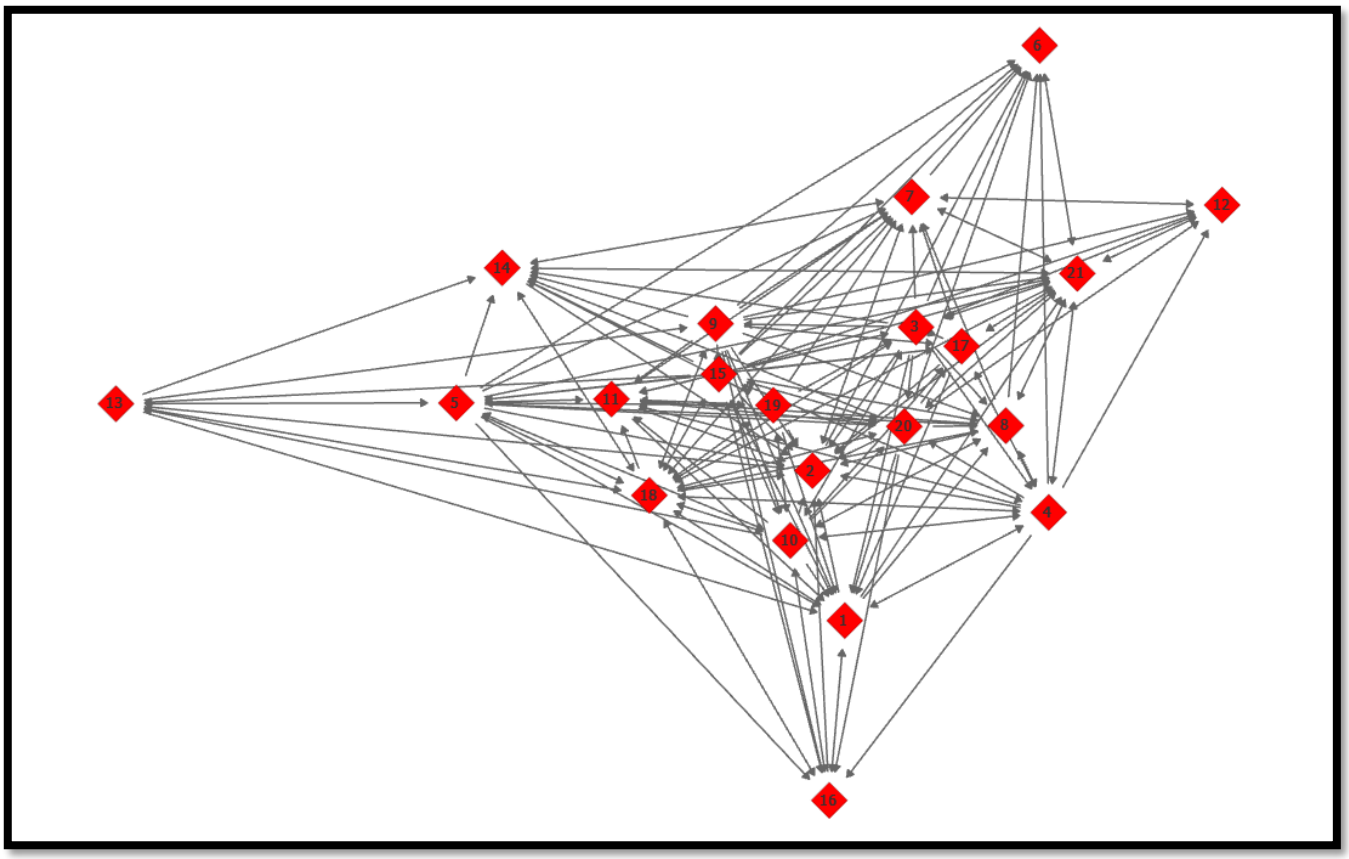
Krackhardt's High-tech Managers is a famous social network of 21 managers of a high-tech US company.
The company manufactured high-tech equipment and had just over 100 employees with 21 managers. David Krackhardt collected the data to assess the effects of a recent management intervention program.

The network consists of 3 relations:

- Advice
- Friendship
- Reports To

Each manager was asked to whom do you go to for advice and who is your friend. Data for the "whom do you report" relation was taken from company documents.

This data is used by Wasserman and Faust in their seminal network analysis book,
Krackhardt D. (1987). Cognitive social structures. Social Networks, 9, 104-134.



Countries' Trade of Basic Manufactured Goods

There is data present on 24 countries and their trade activities with each other.

A plot of the network's adjacency matrix is generated. Each element M_{ij} of the NxN matrix is filled if there as an edge from i to j, and not filled if no edge exists.

A clique consensus report is generated. A clique is the largest subgroup of actors in the social network who are all directly connected to each other (maximal complete subgraph). The Bron–Kerbosch algorithm is applied to produce a census of all maximal cliques in the network.

Maximal Cliques found: 19

Clique No	Clique members
1	14
2	20
3	22 15
4	8 22
5	10 23
6	6 23 3
7	1 24 5 4 13
8	12 2 19 9 13 23 18
9	2 4 19 9 13 23 3 18
10	4 19 9 22 13 23 21 3 18
11	5 4 19 9 22 13 23 3 18
12	11 4 22 13 23 21 17 18 16
13	24 5 19 9 22 13 23 7 18
14	24 11 4 22 13 23 21 18 16
15	24 12 19 9 22 13 23 21 18
16	11 4 19 9 22 13 23 21 17 18
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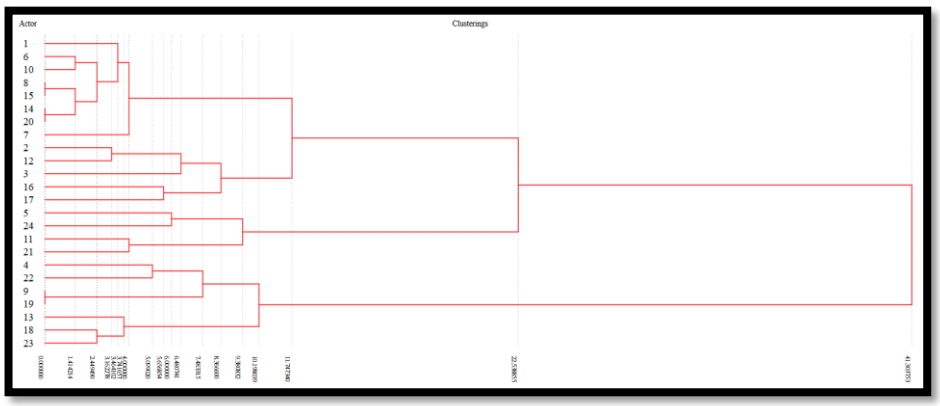
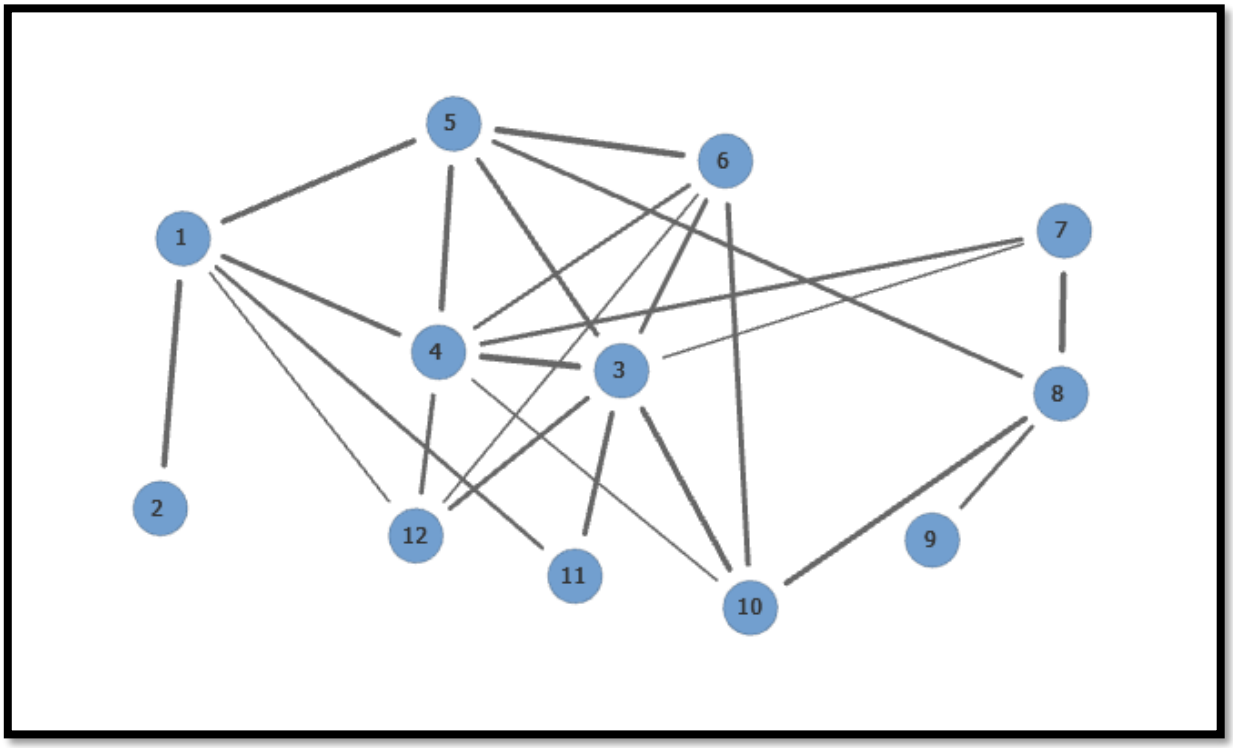
19	24 11 5 4 19 9 22 13 23 18



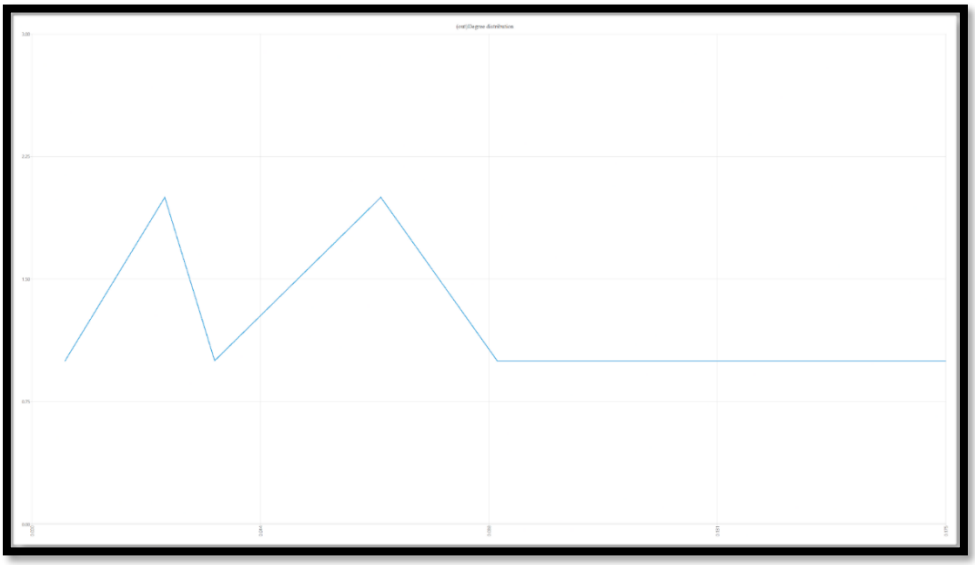
Galada Baboon Colony Network (H22a)

A network of the Galada baboon colony, as described by Dunbar and Dunbar (1975). This is the first set of observations (H22a) and was made on 12 baboons.

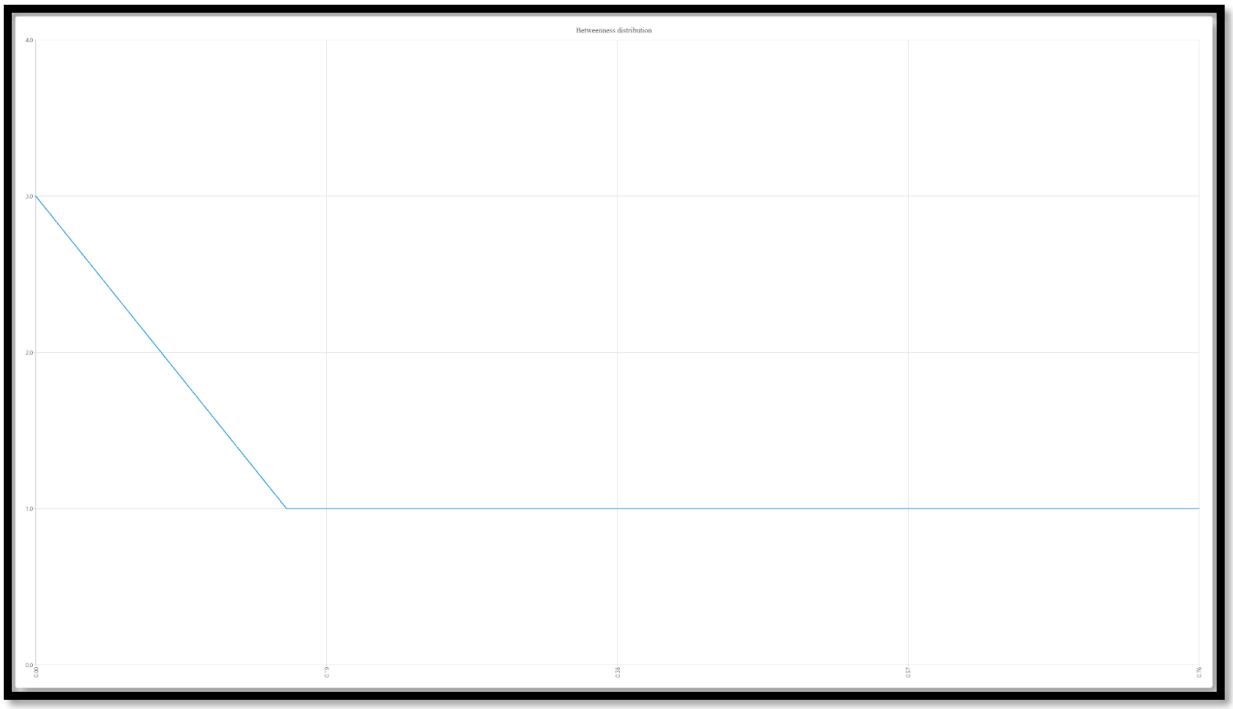
The lines connecting two points (baboons) represent nonagonistic interactions (generally grooming behavior) and the frequency of such interactions is recorded by the edge weight. Data derived from Stephenson & Zelen seminal 1989 paper where they introduced Information Centrality.



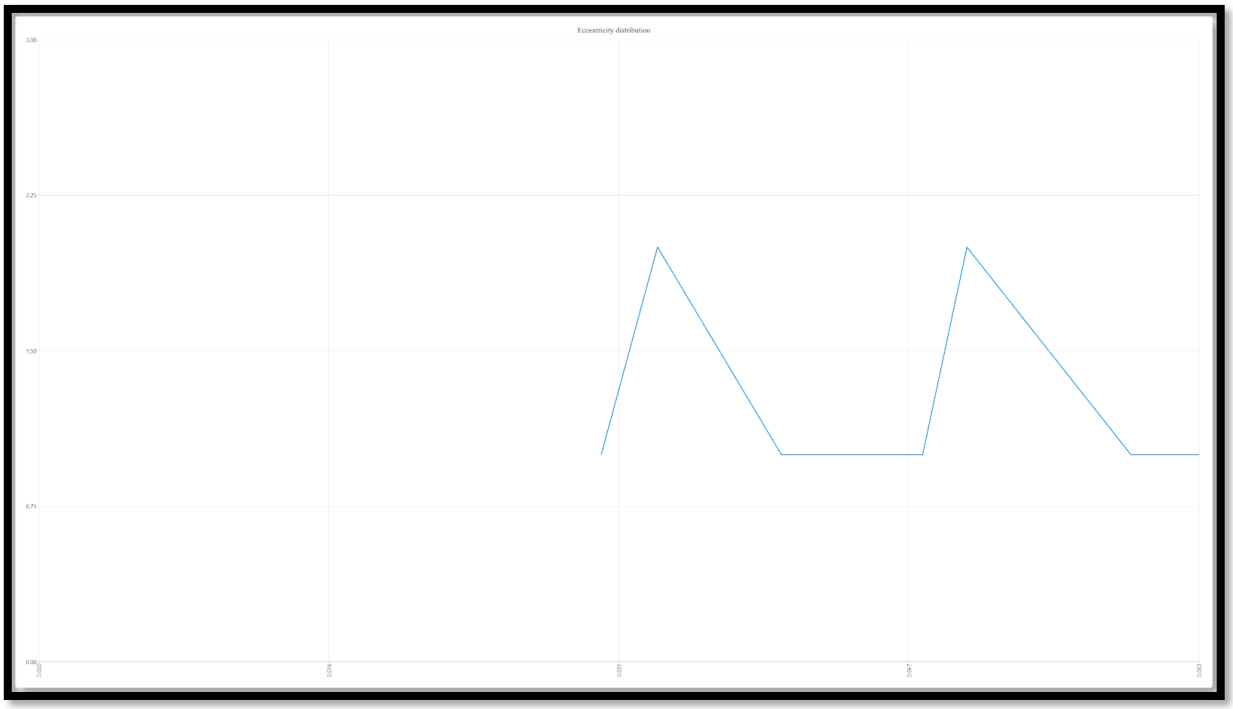
Here, Degree Centrality is computed and visualised.



Then, Betweenness Centrality is computed and visualised.



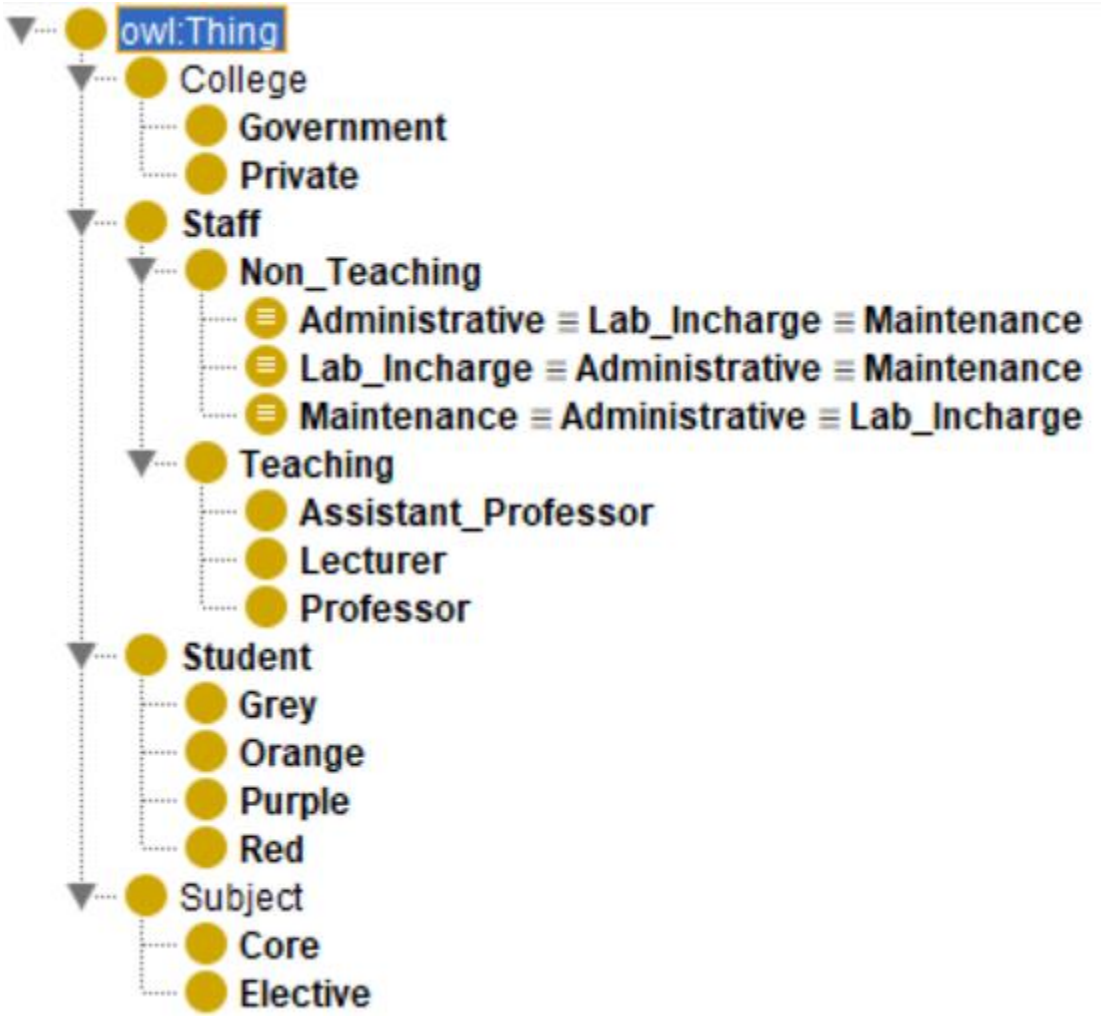
Finally, Eccentricity Centrality is computed and visualised.



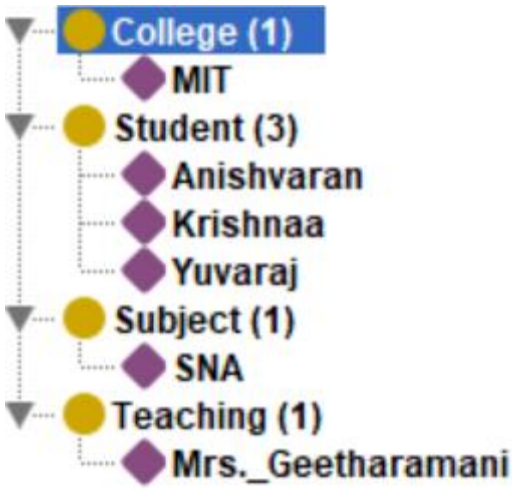
4: Ontology Development

RDF on College

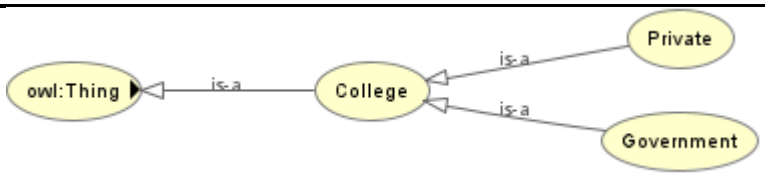
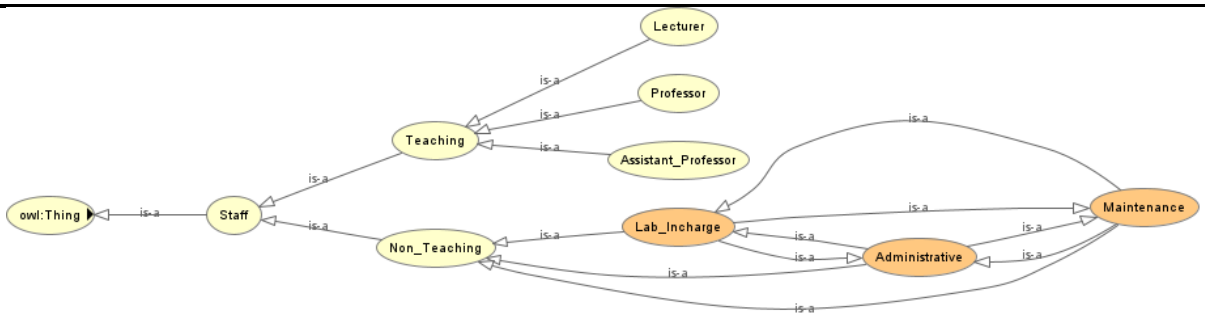
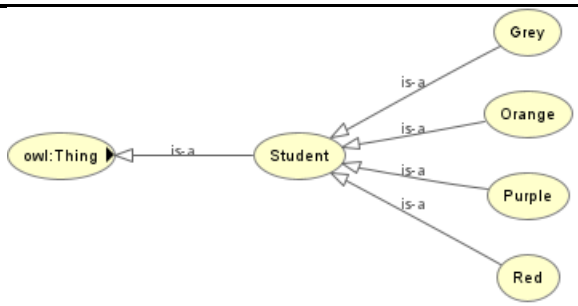
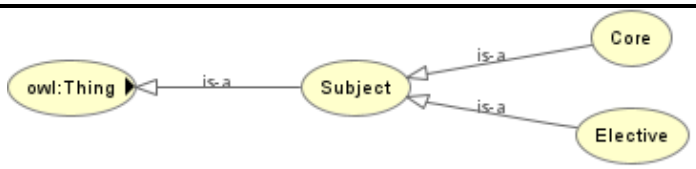
The hierarchical structure for the college ontology is given below.



Individual data has been added.



A visualisation of the constructed ontology is also generated.

Thing (Root)	
College	
Staff	
Student	
Subject